Chapter 20 Problem Set

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1. Let S be the set of all prime factors of N. Finding a factorization of N is the same as splitting S into disjoint subsets. The number of ways to do this for k subsets is $\left\{ {n \atop k} \right\}$, so the total is $\sum_{k=1}^{n} \left\{ {n \atop k} \right\}$.

2. Given a set $[2n]$, you can arrange n of those elements into n different blocks. There are $n!$ ways of doing this. But something counted by $\left\{ {2n \atop n} \right\}$ includes that and the other n elements, so $n! < \left\{ {2n \atop n} \right\}$.

The expression $\left\{ {2n \atop n} \right\}$ counts partitions of $[2n]$ into n subsets. If we rearrange those subsets and the elements in those subsets, we get something counted by $(2n)!$. Therefore, $\left\{ {2n \atop n} \right\} < (2n)!$.

3. We can use complimentary counting for this. If we place 1 in a block of size 1, we just need to partition the rest. So we have to find the number of set partitions of $[n-1]$, which is $\sum_{k=1}^{n-1} \left\{ {n-1 \atop k} \right\}$. Subtracting that from the total number of partitions of $[n]$, we get $\sum_{k=1}^{n} \left\{ {n \atop k} \right\} - \sum_{k=1}^{n-1} \left\{ {n-1 \atop k} \right\}$.

4. If n is odd then the answer is $\left\{ {n \atop 2} \right\}$ since every partition is valid.

If n is even, the number of ways to do this so that the parts are of the same size is $\binom{n}{n/2}$ because we need to choose $\frac{n}{2}$ of the elements to be in one subset. So the answer would be $\left\{ {n \atop 2} \right\} - \binom{n}{n/2}$.

5. We can count the number of ways to have 1 and 2 in different cycles and then subtract it from $\sum_{k=1}^{n} \left[{n \atop k} \right]$. Let's consider a case where there are a elements in the cycle with 1 and b different elements in the cycle with 2. We can choose what the other $a-1$ elements are in $\binom{n}{a-1}$ ways and order them all in $(a-1)!$ ways. That gives us $\frac{n!}{a!}$. From there, we can choose the $b-1$ other elements in the cycle with 2 in $\binom{n-a}{b-1}$ ways and order them in $(b-1)!$ ways. After that, the number of ways to make cycles with the remaining numbers is $\sum_{k=1}^{n-a-b} \left[{n-a-b \atop k} \right]$. Summing over a and b and putting this all together, we get a final answer of

$$\sum_{k=1}^{n} \left[{n \atop k} \right] - \sum_{a=1}^{n-1} \sum_{b=1}^{n-a-1} \frac{n!(n-a)!}{a!b!} \sum_{k=1}^{n-a-b} \left[{n-a-b \atop k} \right].$$

We can use a similar approach to find how many permutations have $1, 2$, and 3 in the same cycle. Let there be a, b, and c elements in the cycles with 1, 2, and 3 respectively. As before, the number of ways to form the cycle with 1 is $\frac{n!}{a!}$, the cycle with 2 is $\frac{(n-a)!}{b!}$, and the cycle with 3 is $\frac{(n-a-b)!}{c!}$. The number of ways to make cycles with the remaining numbers is $\sum_{k=1}^{n-a-b-c} \begin{bmatrix} n-a-b-c \\ k \end{bmatrix}$. Summing over a, b, and c and putting it all together, we get

$$\sum_{k=1}^{n} \begin{bmatrix} n \\ k \end{bmatrix} - \sum_{a=1}^{n-1} \sum_{b=1}^{n-a-1} \sum_{c=1}^{n-a-b-1} \frac{n!(n-a)!(n-a-b)!}{a!b!c!} \sum_{k=1}^{n-a-b-c} \begin{bmatrix} n-a-b-c \\ k \end{bmatrix}.$$

I couldn't figure out how to do the last part of this problem.

8. The LHS is just the number of ways to split $[n+1]$ into $m+1$ subsets. The RHS represents the number of ways to choose k elements out of n, split those elements into m subsets, and sum across all k from 0 to n. For one part of the sum, we are taking k of the n elements and making m subsets from it. The remaining elements can then form their own subset, leaving us with $m+1$ total. To account for the case where $k = n$ and that subset is empty, we can add $n+1$ to it. Therefore, the RHS gives us the number of ways to split $n+1$ numbers into $m+1$ subsets, which is equivalent to the LHS.